SAMPLE

Famous Astronomers: Clyde Tombaugh

Discoverer of Pluto

Aira Astralis

Free sample: Introduction & Chapter 1

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Introduction

The story begins on a Kansas farm with a hailstorm, which is the correct place for it to begin, because the farm and the hailstorm explain everything that follows. A boy who wanted to go to college could not, because the crops his tuition depended on were destroyed in an afternoon, and the boy, refusing to be stopped, built his own telescopes from farm machinery and spare glass, taught himself optics by hand and eye, and mailed his drawings of Mars and Jupiter to an observatory in Arizona as a polite stranger might. The observatory wrote back with a job offer, and within a year the farm kid with no degrees had found the thing the professionals had hunted for a quarter century: a new world at the edge of the planetary system.

The person is Clyde Tombaugh, the American astronomer whose life this account holds, and the framing the series uses for its astronomers fits him more snugly than any other: the person, the era, the discoveries, and the legacy the discipline still carries. He is the only astronomer in this series' shelf, and very nearly the only one in the century's history, to have discovered a planet from a start with no formal training at all, and the account tells that story honestly, with its luck and its errors and its long second act, because the honest version is better than the legend.

The plan follows the series convention: the opening chapters give the life and the machine he worked, the middle chapters the science of his search and the method that found the world, the later chapters the discoveries themselves and their strange afterlife, and the final chapters the small delights, the closing verdict, and the shelf. The middle of the twentieth century is the setting, a time when a hired hand with a good eye and a machine called a blink comparator could outperform every theorist alive, and the account treats that setting with the respect it earns: the era of photographic plates was the last in which patient human attention, unaided, could push the sky's frontier.

The introduction's honesty paragraph is required here more than anywhere, because the subject's famous discovery carries a correction the reader already half knows. The world Tombaugh found was called the ninth planet for three quarters of a century, and then it was not, demoted by a definition the astronomers themselves finally wrote down in 2006, and the account does not treat the demotion as a wound. The truth the discovery carried was stranger and better than the label: the dot Tombaugh caught in 1930 was the first seen member of a belt of small worlds beyond Neptune, the frontier of a region the century's later telescopes filled with thousands of its kind, and the farm boy's careful eye had found the edge of a map no one knew existed. The account keeps both facts, the planet that was and the belt that is, because a biography

that flinches from either fails its subject.

The book's shelf position deserves a sentence here, because the series' gallery of sky people holds farmers and professors both, and this subject sits at one end of the gallery's social range. Clyde Tombaugh is the series' purest self-made astronomer, a farm boy whose first instruments were built from farm equipment, and his planet is the last world in the solar system to be found by a single person looking at glass; the volumes that flank this one tell the stories of doctors' sons and professors' proteges, and the contrast is the point, the sky taking workers from every counter and asking only for the craft. The reader who wants the American version of the astronomical life, the version the textbooks cite as the proof that persistence is a method, holds it here.

One scope note before the story runs, because the subject's record spans two careers and the book keeps them both honest. The planet search's decade is documented in the observatory's own plates and logbooks, and the later decades, the teaching, the missile range, and the professorship, are documented in the universities' records and the subject's own recollections, given generously in interviews across his long retirement; the volatile items, the exact dates of the hiring letters and the anecdotes retold with variations, carry their cautions in the fact notes rather than passing as settled. The reader who wants the discovery's mechanics will find the telescope, the plates, and the comparator treated as characters in their own right, because this biography's truth is instrumental: the farm boy's story cannot be told without the machines he built and the machines that built him.

The account's honest size note: the planet's discoverer published less than the professors of his rank, because his working decades went to the sky and the classroom rather than the journals, and the biography that follows is therefore built from plates, logbooks, and recollection in that order.

Chapter 1: The Discovery of Clyde Tombaugh

The series titles the chapter by its convention, discovery meaning the coming of the person into his work, and Clyde Tombaugh's coming into astronomy is the most American of the shelf's stories, a tale in which weather,

poverty, and stubbornness make a career.

He was born in the first days of February 1906, in a small Illinois town, the first of six children of a farmer, and the family's move to a Kansas farm when he was a teenager set the stage. The farm child's sky is the dark sky, and the boy's interest in the heavens woke early, fed by an uncle's small telescope, a visit to a great observatory in his twelfth year, and the books that found their way to a house of modest means and serious reading. The account notes the

detail every biography keeps: the boy's plans to enter college collapsed when a June hailstorm flattened the family's crops in 1928, the tuition money going with them, and the refusal to surrender took the form of glass.

The account's fairness to the boyhood includes its opportunities, thin as they look. The farm's isolation had its gift, a night sky with no town glow for miles, and the state's schools, the era's mail order courses, and the public library's shelf of astronomy books were enough for a reader with a project, and the project arrived in the mid twenties when the popular magazines printed a reader's sketches of Jupiter's cloud belts beside a description of the amateur telescope that had drawn them. The boy decided to have both, the view and the instrument, and the account's standing detail of the era's prices is the

budget's meaning: a good mirror's cost, in the catalogues, was several weeks of farm wages, and the grinding of one at home, in the barn, with hand tools, cost the winter's evenings, so the winter's evenings were spent, and the farm's own machinery donated the mount.

Tombaugh ground his own mirrors, using farm tools and hand-built rigs, and built reflectors from materials that included, the record faithfully reports, parts scavenged from disused farm machinery. With the finished instruments he made drawings of Mars and Jupiter of a quality that surprised the professionals, and he mailed them to the Lowell Observatory in Flagstaff, asking, in effect, whether they were any good.

The observatory's answer changed his life. The director in Flagstaff, himself a career observer, wrote back with both praise and a job offer, for the observatory needed a pair of steady hands for a grinding, unglamorous project it had promised its dead founder it would finish: the search for a ninth planet. Tombaugh arrived in Arizona at the start of 1929, a young man of twenty two with a farm handshake and no degree, and within thirteen months he had found the world the observatory had hunted since before his birth. The chapter's arithmetic deserves a pause: the professionals had computed and predicted and searched for a quarter of a century; the farm boy, working systematically, found the planet before his first year's contract ended, and the account will show in the method chapter that the result was not luck but a system so patient it could not miss.

The discovery made him briefly the most famous astronomer in the world, and he spent the fame the way the account's later chapters show: quietly finishing a deferred education, discovering thousands more objects in the sky, serving in the war's navigation schools and the missile ranges of the postwar desert, building a university's astronomy program from a converted storeroom, and dying in 1997 with a place reserved for his ashes aboard a spacecraft that

would, eighteen years after his death, photograph the world he had found as a point of light, up close, at the edge of everything.

The chapter's discovery, then, is a compound:

The years between the job offer and the discovery deserve their own paragraph, since the biographies' details carry the era's texture. The new hire arrived at a mountain town of a few thousand people, an observatory on a hill, a shared bedroom, and a salary of a few dollars a day, and his first assignment was to improve the photographic plates' quality, an apprenticeship that meant months of exposure trials and development chemistry before the search could even resume. The account's fairness to the era includes its speed: nothing

about the planet search was fast, from the plates' hours long exposures to the months long waits between field pairs, and the twenty two year old's patience, formed by farm work that also moved in seasons, matched the project's tempo exactly. a person found by an observatory through the mail, a planet found by the person through the plates, and a life found by the work, which is the series' standing note on how astronomy's careers actually begin, not with genius declared but with attention rewarded.

The account adds one paragraph to the chapter's ending, because the convention's word deserves its last defense. To say the person was discovered is to say that the discipline, in rare decades, recognizes what it needs before it knows how to ask for it: the observatory in 1929 did not advertise a post for a genius of the blink machine, it advertised nothing at all, and the letter from Kansas simply arrived at the office of a man tired of answering crank mail and proved, on inspection, to be the best job application the profession ever read, three pages of drawings from a machine built out of farm scrap, and the chapter's discovery closes on the paper's own lesson, that the sky's institutions are wise exactly as often as they read their mail carefully.

The Mars drawings deserve their paragraph, because the hiring that followed them is the story's hinge. Tombaugh,

unable to afford college, did what the era's amateurs did with talent and no diploma: he observed, he drew, and he sent the work to the professionals, sketches of Mars made at his homemade telescope, packed off to the observatory in Flagstaff in the late nineteen-twenties; Vesto Slipher, the director, read the drawings, saw the observer in them, and answered with a job offer, the planet search needing exactly the kind of eye that could render a planet's surface honestly. The Kansas farm thus handed the Lowell Observatory its last great hire by mail, and the chapter's moral is the one the series keeps for such tales: the work spoke first, the credentials followed, and

the sky does not check diplomas.

The farm's own instrument deserves its paragraph, because the homemade telescope is the chapter's emblem and the details are the man. The mirrors were ground from the glass of a broken cream separator and polished by hand over months; the mount was farm timber and repurposed parts; the fittings and the mount came from whatever the farm's shed held; and the finished instruments, several of them, took the Kansas nights and gave back the sky, the planet drawings and the star counts that became the job application. The chapter keeps the emblem honest: the telescope was imperfect by professional standards, and it taught the skills the profession could not, patience, maintenance, and the habit of making the instrument show the truth.

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